# Function Machines

Year 4 · **Mark Scheme**

## Lesson Worksheet — Answers

### Function Basics

**1.** 30

**2.** 23

**3.** 14

**4.** C) 8

**5.** 18 (4 + 2 = 6; 6 × 3 = 18)

**6.** D) 23 (9 × 2 = 18; 18 + 5 = 23)

**7.** 7 (42 ÷ 6 = 7)

**8.** C) 16 (31 − 15 = 16)

**9.** 13 (5 + 8 = 13)

**10.** 45 (9 × 5 = 45)

**11.** A, B and D (4×3=12, 5×3=15, 7×3=21 are correct; C is false since 6×3=18 not 20, and E is false since 8×3=24 not 22)

**12.** 17 (5 × 4 = 20; 20 − 3 = 17)

**13.** 8 (9 + 7 = 16; 16 ÷ 2 = 8)

**14.** D) 26 (10 × 2 = 20; 20 + 6 = 26)

**15.** 0

### Working Forwards & Backwards

**16.** 41 (11 × 3 = 33; 33 + 8 = 41)

**17.** C) 28 (12 − 5 = 7; 7 × 4 = 28)

**18.** 4 (18 ÷ 3 = 6; 6 − 2 = 4)

**19.** C) 7 (33 ÷ 3 = 11; 11 − 4 = 7)

**20.** 12 (15 + 9 = 24; 24 ÷ 2 = 12)

**21.** a = 3, b = 4 (so the rule is ×3, then +4): from 5a+b=19 and 8a+b=28, subtracting gives 3a=9 so a=3, then b=19−15=4; check 8×3+4=28 ✓

**22.** 2 (0.5 × 4 = 2)

**23.** C) 8 (29 − 5 = 24; 24 ÷ 3 = 8)

**24.** 4 (−3 + 7 = 4)

**25.** A, C and E (6→4, 10→12 and 15→22 are correct; B is false since (9−4)×2=10 not 11, and D is false since (12−4)×2=16 not 14)

**26.** 2 (2/5 × 5 = 2)

**27.** 7 (10 + 4 = 14; 14 ÷ 2 = 7)

**28.** C) 6.5 (13 × 0.5 = 6.5)

**29.** 4 (−8 ÷ 4 = −2; −2 + 6 = 4)

**30.** 2 (3/4 + 1/4 = 1; 1 × 2 = 2)

### Multi-Step Challenge

**31.** 26 (7 × 3 = 21; 21 + 5 = 26)

**32.** 11 (20 ÷ 4 = 5; 5 + 6 = 11)

**33.** C) 4 (5 × 3 = 15; 19 − 15 = 4)

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**34.** 24 (9 − 3 = 6; 6 × 4 = 24)

**35.** 21 (6 × 5 = 30; 30 − 9 = 21)

**36.** A, B, C and E are correct (D is wrong — double 10 then take away 5 gives 15, not 16)

**37.** C) 15 (6 × 5 = 30; 30 − 15 = 15)

**38.** 3 (4 + 14 = 18; 18 ÷ 6 = 3)

**39.** 30 (4 + 11 = 15; 15 × 2 = 30)

**40.** 1 (10 ÷ 2 = 5; 5 − 4 = 1)

**41.** C) 7 (20 − 6 = 14; 14 ÷ 2 = 7)

**42.** £27 (6 × £4 = £24; £24 + £3 = £27)

**43.** 17 (4 × 3 = 12; 12 + 5 = 17)

**44.** C) 30 (5 × 5 = 25; 25 + 5 = 30)

**45.** 14 (6 × 3 = 18; 18 − 4 = 14)

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# Homework Worksheet — Answers

## Function Basics

**1.** 56

**2.** 37

**3.** 21

**4.** B) 7

**5.** 22 (6 + 5 = 11; 11 × 2 = 22)

**6.** C) 28 (8 × 3 = 24; 24 + 4 = 28)

**7.** 6 (54 ÷ 9 = 6)

**8.** C) 22 (40 − 18 = 22)

**9.** 20 (9 + 11 = 20)

**10.** 42 (6 × 7 = 42)

**11.** A, C and D (3→12, 6→24 and 7→28 are correct; B is false since 5×4=20 not 18, and E is false since 9×4=36 not 32)

**12.** 14 (4 × 5 = 20; 20 − 6 = 14)

**13.** 7 (12 + 9 = 21; 21 ÷ 3 = 7)

**14.** C) 13 (6 × 3 = 18; 18 − 5 = 13)

**15.** 0

## Applied Functions

**16.** 29 (9 × 2 = 18; 18 + 11 = 29)

**17.** B) 40 (15 − 7 = 8; 8 × 5 = 40)

**18.** 15 (24 ÷ 4 = 6; 6 + 9 = 15)

**19.** B) 8 (28 ÷ 2 = 14; 14 − 6 = 8)

**20.** 7 (11 + 10 = 21; 21 ÷ 3 = 7)

**21.** a = 3, b = 3 (so the rule is ×3, then +3): from 4a+b=15 and 7a+b=24, subtracting gives 3a=9 so a=3, then b=15−12=3; check 7×3+3=24 ✓

**22.** 3 (0.5 × 6 = 3)

**23.** C) 8 (23 − 7 = 16; 16 ÷ 2 = 8)

**24.** 4 (−6 + 10 = 4)

**25.** A, C and E (2→15, 5→24 and 7→30 are correct; B is false since (4+3)×3=21 not 20, and D is false since (6+3)×3=27 not 28)

**26.** 3 (3/4 × 4 = 3)

**27.** C) 9.5 (19 × 0.5 = 9.5)

**28.** 8 (18 + 6 = 24; 24 ÷ 3 = 8)

**29.** 6 (−10 ÷ 5 = −2; −2 + 8 = 6)

**30.** 3 (2/3 + 1/3 = 1; 1 × 3 = 3)

## Standard / Advanced

**31.** 8 (24 − 8 = 16; 16 ÷ 2 = 8)

**32.** 8 (18 + 6 = 24; 24 ÷ 3 = 8)

**33.** 10 (30 ÷ 5 = 6; 6 + 4 = 10)

**34.** 3 hours (£110 − £20 = £90; £90 ÷ £30 = 3)

**35.** 6 (36 ÷ 3 = 12; 12 − 6 = 6)

**36.** 10 (3 + 5 = 8; 8 × 2 = 16; 16 − 6 = 10)

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**37.** C) 5 (£19 − £4 = £15; £15 ÷ £3 = 5 rides)

**38.** 22 (5 × 3 = 15; 15 + 7 = 22)

**39.** C) 32 (10 × 3 = 30; 30 + 2 = 32)

**40.** 14 (16 − 9 = 7; 7 × 2 = 14)

**41.** A, B, C and E are true (D is false — the true value is 20, not 22)

**42.** 5 km (£15 − £5 = £10; £10 ÷ £2 = 5)

**43.** 25 (9 × 2 = 18; 18 + 7 = 25)

**44.** B) 22 (4 + 8 = 12; 12 × 2 = 24; 24 − 2 = 22)

**45.** 9 (first pass: 3×2−1=5; second pass: 5×2−1=9)

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